

SAFETY DATA SHEET

(SDS)

This safety data sheet complies with the requirements of:
Regulation (EC) No. 1907/2006 and Regulation (EC) No. 1272/2008, (EU) No. 2015/830

Revision Date 06-Dec-2023

WAI2 - EGHS - EUROPEAN

Revision Number 3

SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

1.1. Product Identifier

Product Name Nitrate 3 (as NO₃)-HR

Product No AC4007-FOIL
Unique Formula Identifier (UFI) Not applicable

REACH registration number Not applicable

Pure substance/mixture Mixture

Contains Tartaric Acid, Cadmium

1.2. Relevant identified uses of the substance or mixture and uses advised against

Recommended Use Use as laboratory reagent

Uses advised against No Information available

1.3. Details of the supplier of the safety data sheet

Manufacturer, Importer, Supplier Thermo Fisher Scientific
Robert-Bosch-Str. 163505
Langenselbold, GERMANY
Tel.: +49 (6184) 90-6000

E-mail address wlp.techsupport@thermofisher.com

Made in USA

1.4. Emergency telephone number 24 Hour Emergency Phone Number
CHEMTREC®
Within USA and Canada: 1-800-424-9300
Outside USA and Canada: 1-703-527-3887
(collect calls accepted)

SECTION 2. HAZARDS IDENTIFICATION

2.1. Classification of the substance or mixture

Classification - Mixture

Classification according to Regulation (EC) No. 1272/2008 [CLP]

Acute toxicity - Inhalation (Dusts/Mists)	Category 2 - (H330)
Serious eye damage/eye irritation	Category 1 - (H318)
Germ cell mutagenicity	Category 2 - (H341)
Carcinogenicity	Category 1B - (H350)
Reproductive toxicity	Category 2 - (H361)
Specific target organ toxicity (repeated exposure)	Category 2 - (H373)
Acute aquatic toxicity	Category 1 - (H400)
Chronic aquatic toxicity	Category 1 - (H410)

2.2. Label elements

Contains Tartaric Acid, Cadmium



Signal Word

Danger

Hazard Statements

H341 - Suspected of causing genetic defects

H350 - May cause cancer

H361 - Suspected of damaging fertility or the unborn child

H318 - Causes serious eye damage

H373 - May cause damage to organs through prolonged or repeated exposure

H330 - Fatal if inhaled

H410 - Very toxic to aquatic life with long lasting effects

H361fd - Suspected of damaging fertility. Suspected of damaging the unborn child

Precautionary Statements

P304 + P340 - IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing

P320 - Specific treatment is urgent (see supplemental first aid instructions on this label)

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

P280 - Wear eye protection/ face protection

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P310 - Immediately call a POISON CENTER or doctor/physician

P201 - Obtain special instructions before use

P281 - Use personal protective equipment as required

P308 + P313 - IF exposed or concerned: Get medical advice/attention

P202 - Do not handle until all safety precautions have been read and understood

P260 - Do not breathe dust/fume/gas/mist/vapors/spray

P314 - Get medical advice/attention if you feel unwell

P501 - Dispose of contents/container to industrial incineration plant

P273 - Avoid release to the environment

2.3. Other hazards

May be harmful in contact with skin

General Hazards

This product does not contain any known or suspected endocrine disruptors

Toxic to terrestrial vertebrates

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

Component	EC No	CAS No	Weight %	CLP Classification - According to GB-CLP Regulations UK SI 2019/720 and UK SI 2020/1567	REACH Reg. No
Tartaric Acid	EEC No. 201-766-0	87-69-4	30 - 40%	Eye Dam. 1 (H318)	No information available
Sodium Sulfate	EEC No. 231-820-9	7757-82-6	20 - 30%		No information available
Sodium Tartrate	EEC No. 212-773-3	868-18-8	20 - 30%		No information available
CDTA Trisodium Salt	-	36679-96-6	10 - 20%		No information available
Magnesium Sulfate, Anhydrous	EEC No. 231-298-2	7487-88-9	0 - 10%		No information available
Cadmium	EEC No. 231-152-8	7440-43-9	0 - 10%	Acute Tox. 2 (H330) Muta. 2 (H341) Carc. 1B (H350) Repr. 2 (H361fd) STOT RE 1 (H372) Aquatic Acute 1 (H400) Aquatic Chronic 1 (H410)	No information available

Component	CAS No	Specific concentration limits (SCL's)	M-Factor	Component notes
Tartaric Acid	87-69-4	-	-	-
Sodium Sulfate	7757-82-6	-	-	-
Sodium Tartrate	868-18-8	-	-	-
CDTA Trisodium Salt	36679-96-6	-	-	-
Magnesium Sulfate, Anhydrous	7487-88-9	-	-	-
Cadmium	7440-43-9	-	10	-

SECTION 4: FIRST AID MEASURES**4.1. Description of first aid measures**

General Advice	Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.
Eye Contact	Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.
Skin Contact	Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.
Inhalation	Remove to fresh air. If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required.
Ingestion	Do NOT induce vomiting. Call a physician or poison control center immediately.
Self-Protection of the First Aider	Use personal protective equipment as required. See section 8 for more information. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device.

4.2. Most important symptoms and effects, both acute and delayed

Most important symptoms and effects	Causes eye burns, Causes severe eye damage
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4.3. Indication of any immediate medical attention and special treatment needed

Notes to Physician	Treat symptomatically
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SECTION 5: FIREFIGHTING MEASURES**5.1. Extinguishing media****Suitable Extinguishing Media**

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

Unsuitable Extinguishing Media

No information available

5.2. Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors.

5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

SECTION 6: ACCIDENTAL RELEASE MEASURES**6.1. Personal precautions, protective equipment and emergency procedures**

Personal Precautions	Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation. Keep people away from and upwind of spill/leak. Evacuate personnel to safe
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areas.

6.2. Environmental precautions

Environmental Precautions Do not flush into surface water or sanitary sewer system. Vapors may accumulate to form explosive concentrations.

6.3. Methods and material for containment and cleaning up

Methods for Containment Prevent further leakage or spillage if safe to do so.

Methods for Cleaning Up Soak up with inert absorbent material. Pick up and transfer to properly labeled containers.

Reference to Other Sections

Refer to protective measures listed in Sections 7 and 8

See Section 8 for information on appropriate personal protective equipment

See Section 12 for additional Ecological Information

See Section 13 for additional waste treatment information

SECTION 7: HANDLING AND STORAGE

7.1. Precautions for safe handling

Advice on safe handling

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Avoid dust formation. Use only under a chemical fume hood. Do not breathe (dust, vapor, mist, gas). Do not ingest. If swallowed then seek immediate medical assistance.

General hygiene considerations

Handle in accordance with good industrial hygiene and safety practice.

7.2. Conditions for safe storage, including any incompatibilities

Storage Conditions

Keep container tightly closed in a dry and well-ventilated place. Store at room temperature in the original container. Protect from direct sunlight.

7.3. Specific end use(s)

Specific Use(s)

Use as laboratory reagent

Risk Management Methods (RMM)

The information required is contained in this Safety Data Sheet.

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

8.1. Control parameters

Exposure limits

List source(s): **UK** - EH40/2005 Work Exposure Limits, Fourth edition. Published 2020. **IRE** - 2021 Code of Practice for the Chemical Agents Regulations, Schedule 1. Published by the Health and Safety Authority **EU** - Commission Directive (EU) 2019/1831 of 24 October 2019 establishing a fifth list of indicative occupational exposure limit values pursuant to Council Directive 98/24/EC and amending Commission Directive 2000/39/EC

Component	European Union	The United Kingdom	France	Belgium	Spain
Cadmium	TWA: 0.001 mg/m ³ (8h)	STEL: 0.075 mg/m ³ 15 min TWA: 0.025 mg/m ³ 8 hr Carc. metal	TWA / VME: 0.004 mg/m ³ (8 heures). restrictive limit	TWA: 0.01 mg/m ³ 8 uren TWA: 0.004 mg/m ³ 8 uren	TWA / VLA-ED: 0.01 mg/m ³ (8 horas) TWA / VLA-ED: 0.002 mg/m ³ (8 horas)

Component	Italy	Germany	Portugal	The Netherlands	Finland
Tartaric Acid		TWA: 2 mg/m ³ (8)			

		Stunden). AGW - exposure factor 2 TWA: 2 mg/m ³ (8 Stunden). MAK Höhepunkt: 4 mg/m ³			
Cadmium	TWA: 0.001 mg/m ³ 8 ore. Time Weighted Average TWA: 0.004 mg/m ³ 8 ore. Time Weighted Average until July 11, 2027	TWA: 0.002 mg/m ³ (8 Stunden). AGW - exposure factor 8 TWA: 0.002 mg/m ³ (8 Stunden). AGW - Haut	TWA: 0.001 mg/m ³ 8 horas TWA: 0.004 mg/m ³ 8 horas	TWA: 0.004 mg/m ³ 8 uren	TWA: 0.004 mg/m ³ 8 tunteina

Component	Austria	Denmark	Switzerland	Poland	Norway
Tartaric Acid			STEL: 4 mg/m ³ 15 Minuten TWA: 2 mg/m ³ 8 Stunden		
Cadmium	TRK-KZGW: 0.016 mg/m ³ 15 Minuten TRK-KZGW: 0.004 mg/m ³ 15 Minuten TRK-TMW: 0.004 mg/m ³ TRK-TMW: 0.001 mg/m ³	TWA: 0.001 mg/m ³ 8 timer STEL: 0.002 mg/m ³ 15 minutter	Haut/Peau TWA: 0.001 mg/m ³ 8 Stunden	TWA: 0.004 mg/m ³ 8 godzinach	TWA: 0.001 mg/m ³ 8 timer STEL: 0.003 mg/m ³ 15 minutter. value calculated inhalable fraction

Component	Bulgaria	Croatia	Ireland	Cyprus	Czech Republic
Cadmium	TWA: 0.004 mg/m ³	TWA-GVI: 0.004 mg/m ³ 8 satima. applies during the transition period until July 11, 2027 inhalable fraction	TWA: 0.001 mg/m ³ 8 hr. inhalable fraction TWA: 0.004 mg/m ³ 8 hr. limit value 0.004 mg/m ³ until 11 July 2027 inhalable fraction STEL: 0.003 mg/m ³ 15 min STEL: 0.012 mg/m ³ 15 min	TWA: 0.001 mg/m ³	TWA: 0.004 mg/m ³ 8 hodinách. 0.002 mg Cd/g Creatinine in urine inhalable fraction of aerosol Potential for cutaneous absorption Ceiling: 0.008 mg/m ³

Component	Estonia	Gibraltar	Greece	Hungary	Iceland
Cadmium	TWA: 0.004 mg/m ³ 8 tundides. valid until July 10, 2027		TWA: 0.001 mg/m ³	TWA: 0.004 mg/m ³ 8 órában. AK	TWA: 0.001 mg/m ³ 8 klukkustundum. inhalable fraction TWA: 0.004 mg/m ³ 8 klukkustundum. valid until July 11, 2027 inhalable fraction Ceiling: 0.002 mg/m ³ inhalable fraction Ceiling: 0.008 mg/m ³ valid until July 11, 2027 inhalable fraction

Component	Latvia	Lithuania	Luxembourg	Malta	Romania
Sodium Sulfate	TWA: 10 mg/m ³	TWA: 10 mg/m ³ IPRD			
Cadmium	TWA: 0.001 mg/m ³	TWA: 0.004 mg/m ³ inhalable fraction IPRD			TWA: 0.05 mg/m ³ 8 ore

Component	Russia	Slovak Republic	Slovenia	Sweden	Turkey
Tartaric Acid			TWA: 2 mg/m ³ 8 urah inhalable fraction STEL: 4 mg/m ³ 15 minutah inhalable fraction		
Sodium Sulfate	MAC: 10 mg/m ³				
Magnesium Sulfate, Anhydrous	MAC: 2 mg/m ³				
Cadmium	TWA: 0.01 mg/m ³ 1051 MAC: 0.05 mg/m ³	TWA: 0.03 mg/m ³ 8 hodinách manufactured TWA: 0.15 mg/m ³ 8	TWA: 0.004 mg/m ³ 8 urah applies until July 11, 2027 inhalable	TLV: 0.001 mg/m ³ 8 timmar. NGV TLV: 0.004 mg/m ³ 8	

		hodinách others STEL: 0.15 mg/m ³ 15 minútach manufactured STEL: 0.75 mg/m ³ 15 minútach others	fraction	timmar. NGV	
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Biological limit values

List source(s):

Component	European Union	United Kingdom	France	Spain	Germany
Cadmium			Cadmium: 0.005 mg/g creatinine urine not critical Cadmium: 0.004 mg/L blood not critical	Cadmium: 2 µg/g Creatinine urine not critical Cadmium: 5 µg/L blood not critical	

Component	Italy	Finland	Denmark	Bulgaria	Romania
Cadmium		Cadmium: 20 nmol/L urine at the end of a working week; time of day does not matter.			Cadmium: 2 µg/g Creatinine urine end of shift Cadmium: 5 µg/L blood end of shift Protein: 2 mg/L urine end of shift

Component	Gibraltar	Latvia	Slovak Republic	Luxembourg	Turkey
Cadmium		Cadmium: 2 µg/L urine	Cadmium: 3.1 µg/L urine not critical carcinogen, category 2		

Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

MDHS14/3 General methods for sampling and gravimetric analysis of respirable and inhalable dust

Derived No Effect Level (DNEL)

See table for values

Component	Acute effects local (Dermal)	Acute effects systemic (Dermal)	Chronic effects local (Dermal)	Chronic effects systemic (Dermal)
Tartaric Acid 87-69-4 (30 - 40%)				DNEL = 2.9mg/kg bw/day
Magnesium Sulfate, Anhydrous 7487-88-9 (0 - 10%)				DNEL = 21.3mg/kg bw/day

Component	Acute effects local (Inhalation)	Acute effects systemic (Inhalation)	Chronic effects local (Inhalation)	Chronic effects systemic (Inhalation)
Tartaric Acid 87-69-4 (30 - 40%)				DNEL = 5.2mg/m ³
Sodium Sulfate 7757-82-6 (20 - 30%)			DNEL = 20mg/m ³	DNEL = 20mg/m ³
Magnesium Sulfate, Anhydrous 7487-88-9 (0 - 10%)				DNEL = 37.6mg/m ³
Cadmium 7440-43-9 (0 - 10%)			DNEL = 4µg/m ³	

Predicted No Effect Concentration (PNEC)

See values below.

Component	Fresh water	Fresh water sediment	Water Intermittent	Microorganisms in sewage treatment	Soil (Agriculture)
Tartaric Acid 87-69-4 (30 - 40%)	PNEC = 0.3125mg/L	PNEC = 1.141mg/kg sediment dw	PNEC = 0.514mg/L	PNEC = 10mg/L	PNEC = 0.0449mg/kg soil dw
Sodium Sulfate 7757-82-6 (20 - 30%)	PNEC = 11.09mg/L	PNEC = 40.2mg/kg sediment dw	PNEC = 17.66mg/L	PNEC = 800mg/L	PNEC = 1.54mg/kg soil dw
Magnesium Sulfate, Anhydrous 7487-88-9 (0 - 10%)	PNEC = 0.68mg/L		PNEC = 6.8mg/L	PNEC = 10mg/L	
Cadmium 7440-43-9 (0 - 10%)	PNEC = 0.19µg/L	PNEC = 1.8mg/kg sediment dw		PNEC = 20µg/L	PNEC = 0.9mg/kg soil dw

Component	Marine water	Marine water sediment	Marine water intermittent	Food chain	Air
Tartaric Acid 87-69-4 (30 - 40%)	PNEC = 0.3125mg/L	PNEC = 1.141mg/kg sediment dw			
Sodium Sulfate 7757-82-6 (20 - 30%)	PNEC = 1.109mg/L	PNEC = 4.02mg/kg sediment dw			
Magnesium Sulfate, Anhydrous 7487-88-9 (0 - 10%)	PNEC = 0.068mg/L				
Cadmium 7440-43-9 (0 - 10%)	PNEC = 1.14µg/L	PNEC = 0.64mg/kg sediment dw		PNEC = 0.16mg/kg food	

8.2. Exposure controls

Engineering Measures	Ensure that eyewash stations and safety showers are close to the workstation location
Personal protective equipment	
Eye/face Protection	Wear chemical splash goggles and face shield. If splashes are likely to occur: Goggles.
Skin and body protection	Wear protective gloves/protective clothing.
Respiratory Protection	No protective equipment is needed under normal use conditions. In case of inadequate ventilation wear respiratory protection.
Environmental exposure controls	Prevent product from entering drains Do not allow material to contaminate ground water system Local authorities should be advised if significant spillages cannot be contained

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

9.1. Information on basic physical and chemical properties

Physical State	Solid
Appearance	Blue Metallic Powder
Odor	Odorless
Odor Threshold	No information available
pH	3.4
PH Range	1.9-4.9

Property	Values	Remarks • Method
Melting point/freezing point	No information available	
Boiling Point/Range	No information available	
Flash Point (High in °C)	No information available	
Evaporation Rate	No information available	
Flammability (solid, gas)	No information available	
Flammability Limit in Air	No information available	

Upper flammability limit:	No information available
Lower flammability limit:	No information available
Vapor pressure	No information available
Vapor Density	No information available
Specific Gravity	No information available
Water Solubility	Soluble in water
Solubility in other solvents	No information available
Partition coefficient	No information available
Autoignition Temperature	-
Decomposition Temperature	No information available
Kinematic viscosity	No information available
Dynamic viscosity	No information available
Explosive Properties	No information available
Oxidizing Properties	No information available

9.2. Other information

Softening Point	No information available
Molecular Weight	No information available
VOC Content(%)	No information available
Density	No Information available
Bulk Density	No information available

SECTION 10: STABILITY AND REACTIVITY**10.1. Reactivity**

No information available

10.2. Chemical stability

Stable under normal conditions

Explosion Data

Sensitivity to Mechanical Impact	None
Sensitivity to Static Discharge	None

10.3. Possibility of hazardous reactions

None under normal processing

10.4. Conditions to avoid

Extremes of temperature and direct sunlight

10.5. Incompatible materials

No information available

10.6. Hazardous decomposition products

Thermal decomposition can lead to release of irritating gases and vapors

SECTION 11: TOXICOLOGICAL INFORMATION**11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008****Product Information**

Acute Toxicity	No information available
Unknown Acute Toxicity	37.5 % of the mixture consists of ingredient(s) of unknown toxicity.
The following values are calculated based on chapter 3.1 of the GHS document	
ATEmix (oral)	6,886.00 mg/kg
ATEmix (dermal)	2,867.00 mg/kg
ATEmix (inhalation-dust/mist)	0.42 mg/L

Component	LD50 Oral	LD50 Dermal	LC50 Inhalation
Tartaric Acid		LD50 > 2000 mg/kg (Rat)	
Sodium Sulfate	LD50 > 10000 mg/kg (Rat)		LC50 > 2.4 mg/L (Rat) 4 h
Cadmium	LD50 = 1140 mg/kg (Rat)		LC50 = 25 mg/m ³ (Rat) 30 min

Skin Corrosion/Irritation No information available

Serious eye damage/eye irritation No information available

Sensitization No information available

Mutagenic Effects No information available

Carcinogenic effects No information available

Component	EUCARC
Cadmium	Carc. 1B

Reproductive Effects No information available

STOT - single exposure No information available

STOT - repeated exposure No information available

Aspiration hazard No information available

11.2. Information on other hazards

Endocrine Disrupting Properties Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

SECTION 12. ECOLOGICAL INFORMATION

12.1. Toxicity

Ecotoxicity effects

Very toxic to aquatic life with long lasting effects

30% of the mixture consists of components(s) of unknown hazards to the aquatic environment

Component	Freshwater Algae	Freshwater Fish	Water Flea
Tartaric Acid	-	LC50: > 100 mg/L, 96h static (Danio rerio)	-
Sodium Sulfate	-	LC50: = 13500 mg/L, 96h (Lepomis macrochirus) LC50: 3040 - 4380 mg/L, 96h static (Lepomis macrochirus) LC50: > 6800 mg/L, 96h static (Pimephales promelas) LC50: 13500 - 14500 mg/L, 96h (Pimephales promelas)	EC50: = 2564 mg/L, 48h (Daphnia magna)
Magnesium Sulfate, Anhydrous	EC50: = 2700 mg/L, 72h (Desmodesmus subspicatus)	LC50: 2610 - 3080 mg/L, 96h static (Pimephales promelas)	EC50: 266.4 - 417.3 mg/L, 48h Static (Daphnia magna)
Cadmium	-	LC50: 0.0004 - 0.003 mg/L, 96h (Pimephales promelas) LC50: = 0.016 mg/L, 96h (Oryzias latipes) LC50: = 21.1 mg/L, 96h flow-through (Lepomis macrochirus)	EC50: = 0.0244 mg/L, 48h Static (Daphnia magna)

		LC50: = 0.24 mg/L, 96h static (Cyprinus carpio) LC50: = 4.26 mg/L, 96h semi-static (Cyprinus carpio) LC50: = 0.002 mg/L, 96h (Cyprinus carpio) LC50: = 0.006 mg/L, 96h static (Oncorhynchus mykiss) LC50: = 0.003 mg/L, 96h flow-through (Oncorhynchus mykiss)	
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Component	Microtox	M-Factor
Cadmium		10

12.2. Persistence and degradability**Degradation in sewage treatment plant**

Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

12.3. Bioaccumulative potential

Component	log Pow	Bioconcentration factor (BCF)
Tartaric Acid	-1.91	No data available

12.4. Mobility in soil**Mobility**

Component	log Pow
Tartaric Acid 87-69-4 (30 - 40%)	-1.91

12.5. Results of PBT and vPvB assessment

No information available

12.6. Endocrine disrupting properties

This product does not contain any known or suspected endocrine disruptors

12.7. Other adverse effects
Persistent Organic Pollutant
Ozone Depletion Potential

This product does not contain any known or suspected substance

This product does not contain any known or suspected substance

SECTION 13. DISPOSAL CONSIDERATIONS

13.1. Waste treatment methods
Waste from Residues/Unused Products

Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

Contaminated Packaging

Dispose of this container to hazardous or special waste collection point.

Other Information

Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Do not let this chemical enter the environment.

SECTION 14: TRANSPORT INFORMATION

IMDG/IMO

14.1 UN-No UN2810
14.2 Proper Shipping Name TOXIC LIQUID, ORGANIC, N.O.S.
Technical Name Cadmium
14.3 Hazard Class 6.1
14.4 Packing Group III
Description UN2810, TOXIC LIQUID, ORGANIC, N.O.S. (Cadmium), 6.1, III, Marine Pollutant
14.5 Marine Pollutant Not Applicable
Environmental hazard YES
14.6 Special Provisions 223, 274
EmS No. F-A, S-A
14.7 Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code No information available

ADR

14.1. UN number UN2810
14.2. UN proper shipping name TOXIC LIQUID, ORGANIC, N.O.S.
14.3. Transport hazard class(es) 6.1
14.4. Packing group III

ICAO

14.1 UN-No UN2810
14.2 Proper Shipping Name TOXIC LIQUID, ORGANIC, N.O.S.
Technical Name Cadmium
14.3 Hazard Class 6.1
14.4 Packing Group III
Description UN2810, TOXIC LIQUID, ORGANIC, N.O.S. (Cadmium), 6.1, III
14.5 Environmental hazard YES
14.6 Special Provisions A3, A4, A137

IATA

14.1 UN-No UN2810
14.2 Proper Shipping Name TOXIC LIQUID, ORGANIC, N.O.S.
Technical Name Cadmium
14.3 Hazard Class 6.1
14.4 Packing Group III
Description UN2810, TOXIC LIQUID, ORGANIC, N.O.S. (Cadmium), 6.1, III
14.5 Environmental hazard YES
14.6 Special Provisions A3, A4, A137
ERG Code 6L

SECTION 15: REGULATORY INFORMATION**15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture****International Inventories**

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS), U.S.A. (TSCA).

Component	CAS No	EINECS	ELINCS	NLP	IECSC	TCSI	KECL	ENCS	ISHL
Tartaric Acid	87-69-4	201-766-0	-	-	X	X	KE-10801	X	X
Sodium Sulfate	7757-82-6	231-820-9	-	-	X	X	KE-31609	X	X
Sodium Tartrate	868-18-8	212-773-3	-	-	X	X	KE-12382	X	X
CDTA Trisodium Salt	36679-96-6	-	-	-	X	X	-	-	-
Magnesium Sulfate, Anhydrous	7487-88-9	231-298-2	-	-	X	X	KE-22752	X	X
Cadmium	7440-43-9	231-152-8	-	-	X	X	KE-04397	X	-

Component	CAS No	TSCA	TSCA Inventory notification - Active-Inactive	DSL	NDSL	AICS	NZIoC	PICCS
Tartaric Acid	87-69-4	X	ACTIVE	X	-	X	X	X
Sodium Sulfate	7757-82-6	X	ACTIVE	X	-	X	X	X
Sodium Tartrate	868-18-8	X	ACTIVE	X	-	X	X	X
CDTA Trisodium Salt	36679-96-6	-	-	X	-	-	X	-
Magnesium Sulfate, Anhydrous	7487-88-9	X	ACTIVE	X	-	X	X	X
Cadmium	7440-43-9	X	ACTIVE	X	-	X	X	X

Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

European Union

Authorisation/Restrictions according to EU REACH

Component	CAS No	REACH (1907/2006) - Annex XIV - Substances Subject to Authorization	REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances	REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC)
Tartaric Acid	87-69-4	-	-	-
Sodium Sulfate	7757-82-6	-	-	-
Sodium Tartrate	868-18-8	-	-	-
CDTA Trisodium Salt	36679-96-6	-	-	-
Magnesium Sulfate, Anhydrous	7487-88-9	-	-	-
Cadmium	7440-43-9	-	Use restricted. See item 72. (see link for restriction details) Use restricted. See item 23. (see link for restriction details) Use restricted. See item 28. (see link for restriction details) Use restricted. See item 75. (see link for restriction details)	SVHC Candidate list - 231-152-8 - Carcinogenic, Article 57a; Specific target organ toxicity after repeated exposure, Article 57(f) - human health

After the sunset date the use of this substance requires either an authorization or can only be used for exempted uses, e.g. use in scientific research and development which includes routine analytics or use as intermediate.

<https://echa.europa.eu/authorisation-list>

<https://echa.europa.eu/candidate-list-table>

<https://echa.europa.eu/substances-restricted-under-reach>

Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Component	ANNEX I - PART 1 List of chemicals subject to export notification procedure (referred to in Article 8)	ANNEX I - PART 2 List of chemicals qualifying for PIC notification (referred to in Article 11)	ANNEX I - PART 3 List of chemicals subject to the PIC procedure (referred to in Articles 13 and 14)
Cadmium 7440-43-9 (0 - 10%)	i(1) — industrial chemical for professional use sr — severe restriction i(2) — industrial chemical for	i — industrial chemical sr — severe restriction	-

	public sr — severe restriction		
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<https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A32012R0649&qid=1604065742303>.

Take note of Directive 2000/39/EC establishing a first list of indicative occupational exposure limit values

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work

Take note of Directive 94/33/EC on the protection of young people at work

Take note of Directive 94/33/EC on the protection of young people at work. Take note of Dir 92/85/EC on the protection of pregnant and breastfeeding women at work. Take note of Dir 76/769/EEC relating to restrictions on the marketing and use of certain dangerous substances and preparations.

National Regulations

UK - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

WGK Classification

Water endangering class = 3 (self classification)

Component	Germany - Water Classification (AwSV)
Tartaric Acid 87-69-4 (30 - 40%)	WGK1
Sodium Sulfate 7757-82-6 (20 - 30%)	WGK1
Magnesium Sulfate, Anhydrous 7487-88-9 (0 - 10%)	WGK1
Cadmium 7440-43-9 (0 - 10%)	WGK3

Component	France - INRS (Tables of occupational diseases)
Cadmium	Tableaux des maladies professionnelles (TMP) - RG 61,RG 61bis

Component	Switzerland - Ordinance on the Reduction of Risk from handling of hazardous substances preparation (SR 814.81)	Switzerland - Ordinance on Incentive Taxes on Volatile Organic Compounds (OVOC)	Switzerland - Ordinance of the Rotterdam Convention on the Prior Informed Consent Procedure
Magnesium Sulfate, Anhydrous 7487-88-9 (0 - 10%)	Prohibited and Restricted Substances		
Cadmium 7440-43-9 (0 - 10%)	Prohibited and Restricted Substances		Annex I - industrial chemical

15.2. Chemical safety assessment

A Chemical safety assessment according to regulation (EC) No. 1907/2006 is not required

SECTION 16: OTHER INFORMATION

Full text of H-Statements referred to under sections 2 and 3

H331 - Toxic if inhaled

H318 - Causes serious eye damage

H341 - Suspected of causing genetic defects

H350 - May cause cancer

H361fd - Suspected of damaging fertility. Suspected of damaging the unborn child

H373 - May cause damage to organs through prolonged or repeated exposure
H411 - Toxic to aquatic life with long lasting effects
H330 - Fatal if inhaled
H372 - Causes damage to organs through prolonged or repeated exposure
H400 - Very toxic to aquatic life
H410 - Very toxic to aquatic life with long lasting effects

Key or legend to abbreviations and acronyms used in the safety data sheet

CAS - Chemical Abstracts Service

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

NZIoC - New Zealand Inventory of Chemicals

WEL - Workplace Exposure Limit

ACGIH TLV: American Conference of Governmental Industrial Hygienists
- Threshold Limit Value

DNEL - Derived No Effect Level

RPE - Respiratory Protective Equipment

LC50 - Lethal Concentration 50%

NOEC - No Observed Effect Concentration

PBT - Persistent, Bioaccumulative, Toxic

TWA - Time Weighted Average

IARC - International Agency for Research on Cancer

Predicted No Effect Concentration (PNEC)

LD50 - Lethal Dose 50%

EC50 - Effective Concentration 50%

POW - Partition coefficient Octanol:Water

vPvB - very Persistent, very Bioaccumulative

ADR - European Agreement Concerning the International Carriage of Dangerous Goods by Road

IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code

OECD - Organisation for Economic Co-operation and Development

BCF - Bioconcentration factor

TWA TWA (time-weighted average)

Ceiling Maximum limit value

ICAO/IATA - International Civil Aviation Organization/International Air Transport Association

MARPOL - International Convention for the Prevention of Pollution from Ships

ATE - Acute Toxicity Estimate

VOC - (Volatile Organic Compound)

STEL STEL (Short Term Exposure Limit)

Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

Full text of H-Statements referred to under section 3

H330 - Fatal if inhaled

H341 - Suspected of causing genetic defects if inhaled

H350 - May cause cancer if swallowed

H361fd - Suspected of damaging fertility. Suspected of damaging the unborn child

H372 - Causes damage to the kidneys/ liver/ eyes/ brain/ respiratory system/ central nervous system through prolonged or repeated exposure in contact with skin

H400 - Very toxic to aquatic life

H410 - Very toxic to aquatic life with long lasting effects

H318 - Causes serious eye damage

Prepared By	Regulatory Affairs
Prepared For	Thermo Fisher Scientific Inc.
Issue Date	No information available
Revision Date	06-Dec-2023
Reason for revision	SDS sections updated.

This safety data sheet complies with Regulation UK SI 2019/758 and UK SI 2020/1577 as

amended.

Disclaimer

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End of Safety Data Sheet